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CHAPTER 11

Instant Coffee Production

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11.1 Introduction

The oldest records on instant coffee (also known as soluble coffee) date from 1771, in the United Kingdom. However, it was only patented in 1890 in New Zealand, citing the patented "Dry Hot-Air" process, and sold locally named after its inventor, David Strang.[1]

By the end of the 19th century in England the use of liquid or concentrate extracts obtained by batch extraction of ground roasted coffee followed by addition of sugar and vacuum evaporation was common. The first North American similar product was developed around 1853, and L. D. Gale was granted a patent in 1865 to obtain an extract from ground roasted coffee, mixed with sugar. This invention was first shown publicly during the Pan American Exposition of 1901 although about 10 to 20 years before a small-scale trade of dried instant coffee already existed. Only in 1903 an American patent on the process of instant coffee powder production was filed by Satori Kato from Chicago. In 1906, George Constant Washington created "Red E Coffee", the first instant coffee produced on a large scale, introduced to the American market in 1909. During World War I (1914–1918), following the use of ground roasted coffee (adopted as a replacement for rum in wars after 1832) supplied to the troops during the American civil war (1861–1865), instant coffee was regarded as one of the most important articles of

Coffee: Production, Quality and Chemistry
Edited by Adriana Farah
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Published by the Royal Society of Chemistry, www.rsc.org

subsistence, being the entire available production acquired by the army, and peaking at 20 t per day. However, the product was too dark and sticky.[2] In 1930, a Swiss company began to sell instant coffee produced in a battery of percolators and mixed with 50% of corn sugars before drying; they further invented the first freeze-dried instant coffee in 1938 while helping the Brazilian Government solve its problem of green coffee overproduction at the time. During World War II, American soldiers received instant coffee as part of their daily feed ration; the demand was very high (reaching over 13 000 t), promoting the appearance of newcomers to the market. By then, vacuum drying had been replaced by spray drying into towers and around 1950 a North American company began large-scale production with higher yields, obtaining a powdered product of good flow ability without addition of carbohydrates, setting the standard of producing instant coffee exclusively using coffee and water.

In 1966, freeze-dried instant coffee was introduced to the market and a couple of years later instant coffee granules appeared, basically improving the dissolution compared to the powder form and reducing the foam in the cup.[3]

In modern society, being fast and practical is often important; therefore instant coffee consumption has been continuously increasing along with quality improvement. Its production is composed of various unit operations and any thermal treatment affects significantly the product final quality. There are various technological approaches that can be adopted to improve its production and quality; the present chapter will address them.

11.2 Current Uses

Instant coffee is traded in bulk form either as a concentrated liquid extract packed into drums or in its dried forms – powder or granulated, herein packed into cardboard boxes with plastic liners or into super sacks.

It is found in the retail market most commonly in glass or polyethylene (PE)/polyethylene terephthalate (PET) plastic jars, aluminum foiled sachets or tins.

Some consumers drink both roasted and instant coffees, respectively black and under milk/cream, as this last one delivers a rather sweet caramel taste, much appreciated, for instance, in the southern region of Brazil and in the US.

It is used as an ingredient in the manufacturing of dairy beverages, bottled or freshly prepared – very common in Japan and South Korea – and with other dairy products such as ice-creams. It has been available since the 1970s in flavored varieties, with the most recent innovations including instant mixes for latte and mocha beverages and instant iced coffee products with vanilla, mocha and original coffee flavors, mixed with hot milk or boiling water. Some Asian countries commonly consume it as a pre-mix with non-dairy creamer and sugar single doses, known as “coffee mix” or “3 in 1”.[4]

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11.3 Definition

ISO/FDIS 3509 norm defines green coffee as "raw coffee, dry coffee plant seed", roasted coffee as "coffee obtained from roasting green coffee" and instant coffee as "dry product, water soluble, obtained exclusively from roasted coffee by physical methods using water as the only transport agent non derived from coffee".[5] Thus, instant coffee is the product of the aqueous extraction of roasted and ground coffee beans.

Along with its intrinsic definition, best manufacturing instant coffee practices may apply various international norms, such as acrylamide (ISO/CD 18862:2014), authenticity (ISO 24114:2011), caffeine (ISO 10095:1992), carbohydrates (ISO 11292:1995), density (ISO 8460:1987), moisture (ISO 3726:1983, ISO 20938:2008) and sampling (ISO 6670:2002).

Many of the presently available accreditations for the food industry together with some specific programs for coffee have been adopted by various instant coffee manufacturers such as ISO 9000, ISO 14000, ISO 18000, ISO 22000, BRC, 4Q, UTZ, organic, Fairtrade and Rainforest.

11.4 Production

Worldwide overall production of freeze dried instant coffee is currently estimated to be 200 000 t year^{-1} while the production of powder or granules is in the range of 500 000 to 900 000 t year^{-1}.[6] Most industrial plants have the instant coffee production process as automated as possible, with data collection throughout the steps for quality and performance controlling.

Final product controlled attributes further comprise color, mycotoxins such as ochratoxin A, particle size distribution, sediments, carbonized or scorched particles, flow ability, microbiological and sensorial requirements, metal particles, pesticides, *etc.*

The flowchart presented in Figure 11.1 contains the different steps involved in instant coffee production.

Figure 11.1 also highlights possible unit operations throughout the processing where volatile components might be recovered and on the other hand processing points in which supposedly they might be added back. As can be observed, these might happen in parallel. For example, reincorporating aromas into the concentrated coffee extract would proportionally augment those volatiles retention during drying, therefore enhancing the volatiles content in the final product and leading to a superior flavor quality.

11.4.1 Green Coffee

The manufacturing of instant coffee begins with the selection of suitable raw materials, generally variable blends of *C. arabica* and *C. canephora* (robusta or other variety) beans from the various producing regions comprised of the so-called coffee belt to deliver the final desired taste

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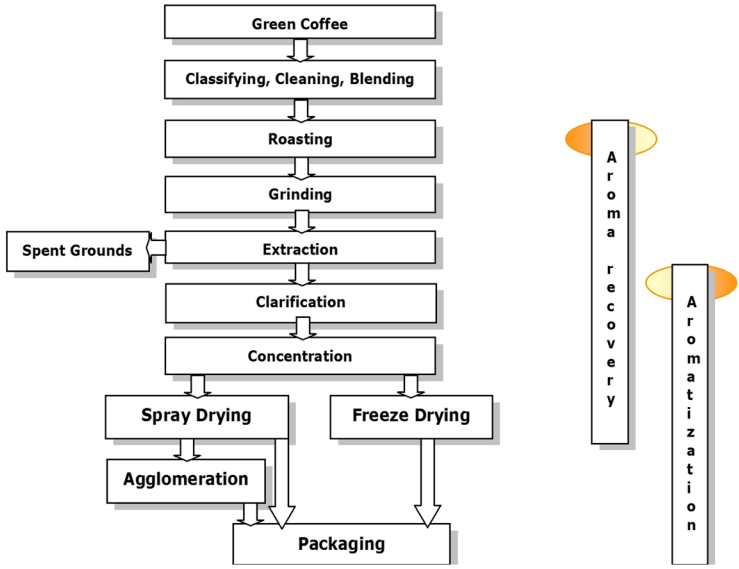


Figure 11.1 Simplified flowchart of the industrial processing of soluble coffee.

attributes. Predominantly, coffees from *C. canephora* species represent the most used beans since they deliver a higher overall yield, as detailed in Section 11.4.4.1.

Grading, storing and blending of the green coffees must observe the best manufacturing practices and controls; de-stoning is relevant to avoid any contamination and eventual broken or faulty equipment. On the sensory side, similarly to roasted coffee, black-green-stinker beans (known as coffee defects) are also crucial to the instant coffee final flavor and must be kept under strict control for better quality.

Typically, green or roasted beans are conveyed *via* bucket elevators, pneumatic transportation and or chain/belt conveyors inside the instant coffee manufacturing plants.

11.4.2 Roasting

Once selected, the beans are roasted, basically following the processes described in Chapter 8 mainly applying similar equipment to normal roasted coffee production. Here again the roasted beans undergo de-stoning and removal of eventual large metal particles (ferrous and non-ferrous), which is important for both product and equipment safety.

Once roasted, the beans might be degassed. Overall, standard roasting conditions lead to a high content of carbon dioxide (CO_2) produced during the roasting operation, mainly linked to the carbohydrates degradation, in addition to other compounds – approximately 10 liters of CO_2 per kilogram of roasted whole beans are produced.[7,8] Unlike in ground

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roasted coffee production where the CO_2 mainly affects packing, in the case of instant coffee, the CO_2 present will be directly linked to issues like forming thick and resilient foam on the extracts, eventually leading to final powder low density and light particle colors. The CO_2 release from the roasted beans follows a first-order degree reaction, and the right degassing period before extraction will help to avoid gas plugging on the pipelines and reduce foam, preventing relevant flavor losses or changes. The coffee aroma compounds are also released in accordance with a first-order degree reaction, but with a slower velocity when compared to the gases release mentioned above.[9]

Coffee aroma is a highly complex balance of diverse compounds impacted by the presence of water and temperature.[10]

11.4.3 Grinding

Once roasted the beans are ready to move forward into grinding, normally using high capacity roller mills or equivalent coupled or not with particles classifying devices. The desired particle size distribution, generally coarser than the one used to brew a French press coffee (typically coarse), will impact the behavior of the water flow through the coffee bed. For instance during extraction it may favor channeling formation and/or flow plugging as well as excessive pressure drops and presence of fines in the coffee extract, therefore directly linked to a smooth and steady process flow and extraction reaction yields.

11.4.4 Extraction

Industrial roasted coffee extraction is normally performed by percolating water through the ground coffee beans. The water must be potable or treated, in some cases softened – which means reducing the total content of mineral solids present, particularly calcium; eventually demineralized water can be applied.

The water is pumped sequentially through an array of multiple reactor vessels, "washing" the coffee beans, until reaching the total extraction of the soluble solids under the applied conditions; as temperatures at points would allow water ebullition, the reactors are pressurized in a way to always maintain liquid state throughout the process, favoring and optimizing the soluble solids extraction.

The reactors, referred to as extraction columns, normally are in a battery conformation set-up constituted of a series of 3 to 12 interconnected columns, filled up with ground coffee beans. In order to keep the process running continuously, a couple of cells already cleaned, cooled and loaded with fresh degassed and coarser ground coffee are kept waiting ready to undergo percolation through the first-in first-out process.[11]

The percolation battery normally characterizes a semi-continuous or batch counter-current process, where normally the coffee bed remains static

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Figure 11.2 Extraction columns formats.

into each column and the water flows through the battery while changing the thermal load applied to each reactor accordingly.

The extraction columns may be arranged in a straight line or in a circular/semicircular way – for a more compact design; they might be more than 6 meters high and hold from a few hundred up to over a thousand kilograms of roasted and coarse ground roasted coffee. Their adopted geometry varies, such as "tall" form – ratio between the column length (L) and its inner diameter (D) up to 12 – or "short" form – L/D ratio might be as low as 1 (Figure 11.2).[11] The last ones will allow finer coffee bed grinds and faster extraction times and thus less heat damage, generally yielding better flavor – with a relatively lower yield as a setback.

On each extremity the columns are equipped with metallic filters or strainers to hold the ground coffee beans inside the column, yet allowing the aqueous coffee extract to flow through the openings.

The coarser ground roasted beans might get moistened before being transferred into the extraction columns, which will help the beans accommodation inside the column and settle down eventual fines and dust released during grinding. They are normally screw or chain conveyed fed into the extraction column from its top; in some installations a slight vacuum is exerted to the column bottom aiming for better coffee bed compaction.

There are various techniques described to extract coffee beans like "one feed water stream" and "two feed water streams", for example. In the "one feed water stream" technique there may be a conventional extraction (with a single draw-off stream) or a split extraction (the freshest initial portion of the extract is weighed separately, followed by a second part of more extracted coffee). The hottest water passes through several "hot" cells – from the hottest to the coldest with reaction temperatures typically between 180 and 140 °C.[12] The "hot" cells contain the most extracted/spent ground beans, which will be discarded sooner. The liquid flow then passes through two or more "cold" cells – inlet temperatures around or below 100 °C for extraction of the more flavorful compounds. The "cold" cells have the freshest or less extracted coffee beans.

In the "two feed water streams" technique, known as double extraction, the first extraction is performed in a set of columns with temperatures typically

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around 100 °C where the fresh coffee is, with a secondary extraction performed using water at temperatures up to 180 °C reaching the older columns with the coffee previously extracted in the first extraction. As a rule, the hot water enters the set at the most extracted cell and the extract leaves at the less extracted one where the fresh roasted beans are, in a way that improves the overall aroma and taste of the product.

The coffee extract is collected from the coolest end, immediately being thereafter cooled down below 15 °C, and weighed for process controlling purposes.

The extraction process comprises various critical variables, such as: extraction cycle time, number of on-line extraction columns, quantity of extract collected in every cycle named draw-off, temperature of each extraction column and the temperature profile throughout the entire battery. Typical conditions would be for extraction cycle times ranging from 15 to 50 minutes.

The higher the water to coffee ratio used the more acid the extract and the higher the expected yields. The use of various temperature ramps with intercooling or heating might be applied to the extraction set. The combination of the above variables will ultimately define the coffee extract flavor quality and the extraction yield – the content of soluble solids obtained from the roasted and ground coffee and the solids concentration reached in each batch. It is also possible to use continuous counter extraction processes, driven by screw conveyors that move the fresh coffee towards the direction of the hotter temperatures. These are more applicable when lower yields are envisaged.

Soon after extraction, the coffee extract is a dark brown color coffee-flavored liquid with a content of soluble solids ordinarily ranging from 10% up to 20%,[11] depending upon the processing conditions, being naturally slightly acid – typical pH range from 4.7 up to 5.4.[13] Therefore, from this unit operation onwards all the installation in contact with the coffee extract (pipelines, pumps, tanks, equipment, *etc.*) must be of sanitary grade stainless steel, allowing the necessary Clean-in-Place (CIP) systems efficiency as well as fulfilling Hazard Analysis and Critical Control Points (HACCP) requirements. Despite this, the coffee extract is not a good microbiological growth medium; on the contrary it tends to reduce contamination with time (due to its acidity, presence of some compounds with antimicrobial activity like caffeine and chlorogenic acids, for example).[13–15] However, biofilm formation and mold development occur and are the main points of attention.

When the oldest cell in the battery is considered wasted, it is isolated from the process and the coffee spent grounds are discharged, normally by releasing the pressurized column contents and collecting them into cyclones under lower pressure or atmospheric conditions. The emptied column is then ready to re-start into the percolation cycle.

The collected wet spent grounds (moisture around 80%) in general are mechanically expelled to reduce its moisture content to around 50% and

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then burned as a renewable energy source in order to generate steam, used as fertilizer upon further composting processes or as livestock feeding after composting.[16–18] Due to the high oil content (above 20%–25% in dry basis), the dry spent grounds have been recently considered for the production of biodiesel.[19,20]

11.4.5 Extract Clarification

The aqueous extract normally undergoes a filtering step that may be performed by using either centrifuges or filters, targeting the removal of insoluble pieces of the coffee beans that could have passed through the extraction columns filters and any other insoluble particles formed inside the percolators themselves. The obtained extract is often called clarified extract.

Additionally, a desludging decanter step might be adopted as well in order to recover more of the coffee solids from the discharged sediments obtained from normal centrifuges, maximizing yields by adding them back into the clean extract flow.

11.4.6 Extract Concentration

Considering either a spray or freeze drying process, in general the higher the infeed concentration the higher is the volatile compounds retention and there is less product exposition to thermal damage. Due to this and also considering cost constraints, the clarified coffee extract is treated in one or more of several ways to increase its concentration, targeting from 25% up to 60% of soluble solids concentration.[21] In most cases the extract is thermally evaporated normally under vacuum, using multiple stages falling film concentrators with thermal or mechanical vapor recompression, plate and frame evaporators, compact or centrifugal thermal evaporators.

In freeze concentration, the extract is slowly cooled down to freeze out pure ice crystals that are subsequently mechanically removed from the coffee concentrate. This leads to very high quality extracts, although reached concentrations are around 30% to 35% soluble solids, comparatively smaller than when applying thermal concentration.

Reverse osmosis or membrane filtration systems, crystallizers and infrared techniques might also be applied as alternative techniques to concentrate coffee extracts.[22]

Depending on the concentration and composition, the concentrated coffee extract (or thick extract) may present a high viscosity that eventually may hinder further processing and pumping, thus limiting the operational concentrations to equivalent viscosities around 0.3 Pa s.

Once the concentration is completed, the concentrated coffee extract might be commercialized and applied in vending machines or as an ingredient in beverages.

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11.4.7 Aroma Recovery

Among various employed techniques, gas chromatography coupled to mass spectrometry and olfactometry (GC-MS/O) has been applied to roasted coffee with more than 1000 volatile compounds identified, positioning the coffee beverage as a high complexity drink.[23]

Generally speaking, several volatile components present in foodstuffs have little sensorial relevance and recent methodologies applied have made possible to researchers the identification and quantification of a comparatively reduced number of key aroma components, decisive of the odor finally sensed. In coffee, only a small percentage of volatile compounds is responsible for its typical aroma perception.[24,25]

At the end of the extraction and concentration processes, the total concentration of aroma impact compounds in instant coffee tends to be lower than in the original ground roasted coffee.[26] Summarizing, the main contributing factors for this would be: the non-volatile matrix present along with the volatile compounds in roasted and ground coffee, whose interactions would take a relevant role on the volatiles release; the amount of non-volatile soluble material that has been concentrated, whose concentration is around two and a half times higher for instant coffee than in ground roasted coffee; the losses of volatile compounds and further thermal treatments to which instant coffee is subjected during the several steps of the manufacturing process.[27,28]

Water is of utmost importance when extracting coffee, as it significantly influences extraction yield and flavor. Its mineral composition will impact the beverage brightness and, especially when consuming instant coffee with milk, it might significantly affect the color hue based on the ions present (for instance, particular concentrations of iron ions would turn the beverage color towards a different shade of red brown).

On the other hand, there are many opportunities for volatile compounds recovery along with the instant coffee manufacturing process itself. For instance, roasting or grinding gases can be trapped using cold temperatures or cryogenic techniques and be collected; ground, roasted coffee can be heated, steam stripped or solvent extracted or have part of the volatiles extracted using supercritical carbon dioxide; prior to extraction, aromatic coffee oil might be expelled or extracted with alcoholic solutions or liquid carbon dioxide; volatiles can be recovered from coffee extract during or after extraction, or collected from the water removed during concentration and then properly condensed.[29]

By selecting or combining the above-mentioned volatile fractions that eventually might be further processed for enrichment, these volatile compounds might be added back at a later step to try to maintain the aroma as close to the original as possible and produce an attractive and special instant coffee product – for example, these might be added back to the concentrate extract just before drying or perhaps onto the final product or added back directly in the final packaging. Special care might be observed for aromatized instant

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coffees such as protection from light and control of oxygen levels present in the inner atmosphere, preventing degradation and oxidation reactions.

11.4.8 Drying

Just before entering the so-called dry area, the concentrated coffee extract might be filtered, eliminating virtually all impurities present on the product while in the liquid state that may even come from the process itself, like metallic particles.

SPRAY or FREEZE drying are the two basic unit operations normally used for removing the final portion of water from the concentrated coffee extract until a powder around less than 5% moisture is achieved. Although contact times in spray drying are typically under one minute, it is done at higher temperatures which affect the taste of the final product, imparting caramel notes to it. It is less costly than freeze drying though, which uses high vacuum and longer residence times (up to 5 hours). Freeze drying would result in a higher quality product, under the same comparative basis.

In both processes metal detection and removal are important parts of the HACCP practices. Commercial instant coffee would have a compacted density from 220 to 240 kg m^{-3}, conforming to standard jars volumes worldwide.

Particularly applicable to freeze dried or agglomerated instant coffee, a common practice is to apply a fine atomized layer of roasted coffee oil onto the final dried product at a final content up to 0.3%; this will reduce the dust content allowing cleaner inner walls when the package is a transparent one with better product appearance.

The glass transition temperature, which can be defined as the temperature at which an amorphous system changes from the glassy to a rubbery state, is very important to be taken into account on both processes. Typically, dried instant coffee tends to start melting around 60 to 70 °C – also known as sticky temperature – and might adhere to surfaces and begin to create lumps.[30]

If instant coffee is exposed for longer periods to temperatures around those, it might begin such reactions – re-crystallization of sugars is relevant here as well. As the reactions are exothermal they will be self-sustaining an ongoing process for several days, even reaching product temperatures above 45 °C, forming big chunks of product on the already packed instant coffee.

Therefore, it is of utmost importance to cool down the instant coffee as soon as it leaves the drying chambers to product temperatures lower than 30 °C by adding dehumidified cold air.[11]

11.4.9 Spray Drying

Cooled, clarified liquid concentrate coffee extract is now ready for drying until reaching a moisture content under 5%. The liquid extract is atomized into small droplets to allow better heat and mass transfer, being sprayed through a nozzle or rotary atomization device at the top of a drying tower

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Figure 11.3 Pressure atomization.

of various shapes and where inlet temperatures, extract concentration and extract composition allied to the type of nozzle and operating conditions will define the particle size distribution, density and color.

Various models of pressure nozzle atomization (Figure 11.3) can be used, each having its own advantages and disadvantages, as well as rotary atomizers (Figure 11.4) that would lead to higher processing volumes but requiring a wider chamber avoiding the heavier droplets to reach the drying chamber walls.

As can be seen in Figure 11.5, filtered heated inlet air up to 300 °C is typically blown downwards co-currently along with the mist of coffee extract atomized particles to evaporate the water, although counter current drying might also be practiced. The air is diverted out of the tower near the bottom at around 110 °C, together with the evaporated moisture, having its finer particles and dust removed into a cyclone or bag filter. The dry soluble matter that is collected at the bottom of the tower constitutes instant coffee powder – typically spherical brownish particles about 300 μm mean size.[11,31] The beads conformation will depend on the combination of variables like: inlet air temperature, turbulent or steady flow inside the drying chamber, total residence time inside drier, type of nozzle (orifice, chamber, design), nozzle positioning, number of nozzles used, atomization pressure, extract concentration and extract intrinsic composition (linked to surface tension and viscosity), extract inlet temperature, *etc.* This will affect the drying, yielding thicker or thinner crusts, porous or not, round or shriveled or exploded particles that ultimately will modify the powder density, color, particle size distribution and fines content as well powder flow ability.[32,33]

Spray drying may be followed by a further process step to build the powder into coarser particles that will dissolve more completely and readily in

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Figure 11.4 Rotary atomization.

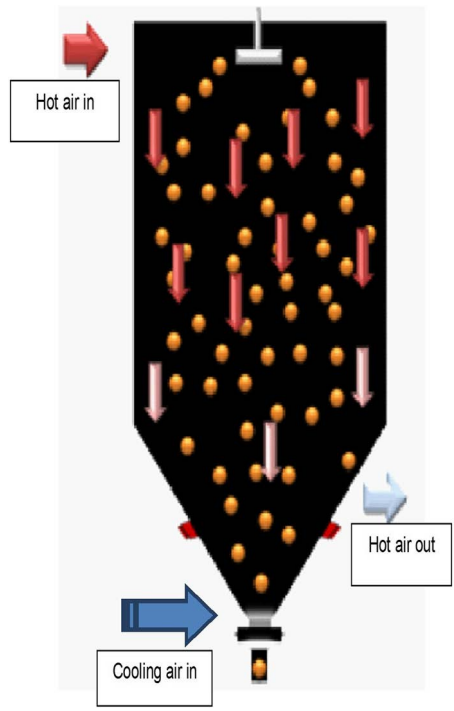


Figure 11.5 Co-current spray drying.

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the consumer's cup. The agglomeration process basically takes further finer ground powder beads together with steam and heat-fuses them forming larger and more granular particles. This is accomplished by exposing the powder to steam or a fine mist, while tumbling it in the air, generally industrially performed either in a tower type equipment or horizontal tunnels called agglomeration units.

11.4.10 Freeze Drying

The freeze drying process involves four steps, beginning with chilling and "foaming" the concentrated coffee extract into a slushy, sorbet-like consistent fluid at about −6 °C, variable according to the extract composition and viscosity.

Then it is spread onto a steel belt or drum or placed into trays and further deep "freezing" until it reaches temperatures of −40 to −45 °C, now a solid block or slab. Quick initial cooling processes result in smaller, lighter colored products, while slower processes would generate larger, darker granules; this is directly linked to the ice crystals growth and melting equilibrium.

The thus formed ice chunks or slabs are hammered broken and "sieved" classified into particles of the proper size for the drying step – this is where the product final shape comes from, like brittle crushed caramelized sugar; excessive fine particles might be re-melted and reworked.

The frozen particles, often placed into special aluminum trays, are sent into a "drying" tunnel (Figure 11.6) – either continuous or batch type, where under high vacuum – typically 6600 Pa – they are submitted to sequential



Figure 11.6 Batch freeze drier.

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heating zones defining a specific temperature profile for faster or more gentle drying. The water sublimates out, being condensed and removed, with the total drying process generally taking around five hours. After leaving the tunnel, the final product is then cooled down, sieved and finally packed. Control of metallic particles is carried out, since the product might get contaminated during the handling into trays for instance with aluminum, mostly used due to its heating conductivity properties.

Other instant coffee drying processes might be considered as well, such as vacuum drum drying, flash drying, rotary and fluid bed driers as well as infrared drying.

11.5 Packaging

Due to its intrinsic composition and the dried particles conformation (diameter, thickness, porosity, *etc.*), instant coffee might be very hygroscopic, quickly absorbing moisture from the air and eventually lumping and coalescing becoming liquefied. Consequently, it must be packaged under low humidity conditions in a controllable moisture container to keep the product dry until purchased and opened by the consumer. Also, to prevent loss of aroma and flavor and allow suitable product shelf-life, the inner atmosphere might be modified by either a reduced oxygen level usually achieved by flushing the headspace with carbon dioxide or nitrogen gaseous streams or applying partial vacuum.

11.6 Decaffeination

Commercial scale decaffeination of instant coffee almost always happens at the green coffee bean level, before the critical roasting process – which will determine the coffee's flavor and aroma – takes place, similarly to roasted and ground decaffeinated coffee production. Therefore, in this case the processing follows the normal conditions described in this book. Caffeine might also be removed directly from the coffee extract itself, widening the process flexibility.

Whichever is the decaffeination process followed, the overall consumption is not high and typically the same instant coffee processing facility would produce normal and decaffeinated instant coffees; thus, monitoring the caffeine content is very important to ensure in process segregation of the decaffeinated instant coffee production batches from the "non-decaffeinated" flows. Decaffeination processes are described in Chapter 7 of this book.

11.7 Trends

Instant coffee was brought about because of both the need of the coffee industry to expand the consumption itself and, on the other hand, the consumers' desire for such a hot beverage to be readily available. It is very

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convenient and practical to prepare, and can be considered as the opening door to new and emerging markets, such as China, where the traditional beverage is tea that presents a milder taste profile, more aligned with that of instant coffee.

In recent years, overall coffee consumption is growing in emerging countries as well as in coffee-producing countries and one in three cups of coffee worldwide is being made out of instant coffee. Considering instant coffee process yields, this is equivalent to approximately 20% or more of the world's green coffee beans production.[34]

The development of new technologies, such as aroma recovery, has facilitated the creation of new types of instant coffees with more coffee flavor, reaching in some cases quality levels comparable to brewed coffees; gourmet instant coffees such as Carte Noire® or VIA® product line – launched in 2009 – have revolutionized the instant coffee scenario with prospects of increased instant coffee consumption already evident in emerging countries and Asia. Overall coffee consumption is forecast by experts to increase up to 14 million bags of 60 kg in the next 10 years, and at least half of such an increase will come from instant coffee, with an expected 4% growth for instant coffee until 2017.[34,35]

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